Mechanical Engineering Experiments (II) — Report 3

機械工程實驗(二) 第三次報告

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Q1:

(a) .

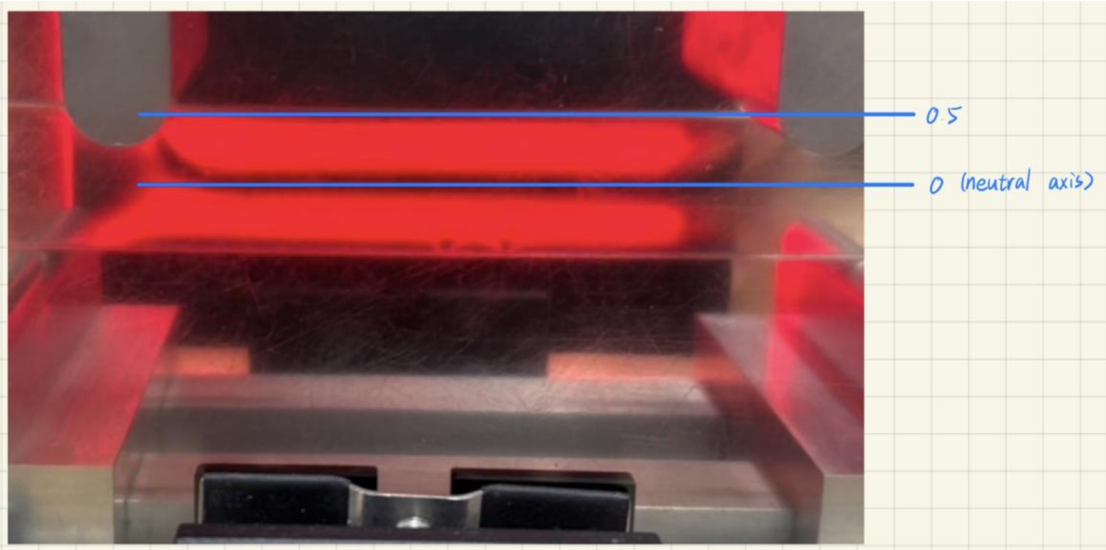

Figure 1. Neutral axis and stripe numbers for $m = 1\ kgw, l = 2\ cm$

(Since calculating the error for Q2 becomes meaningless if N = 0, a value of 0.5 is used instead)

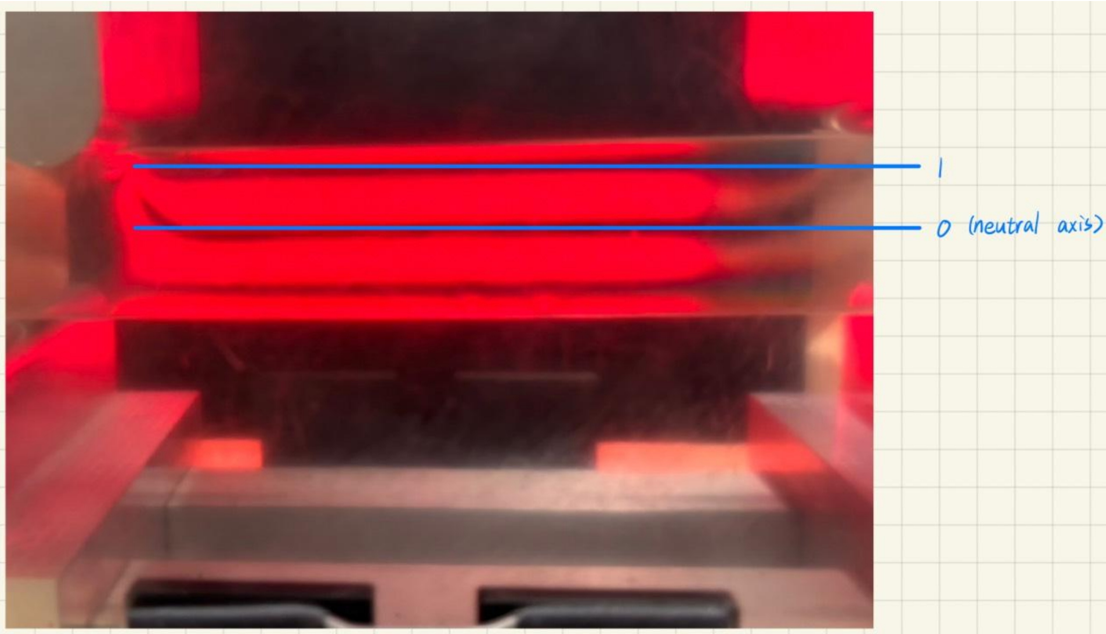

Figure 2. Neutral axis and stripe numbers for $m = 2.5\ kgw, l = 1.7\ cm$

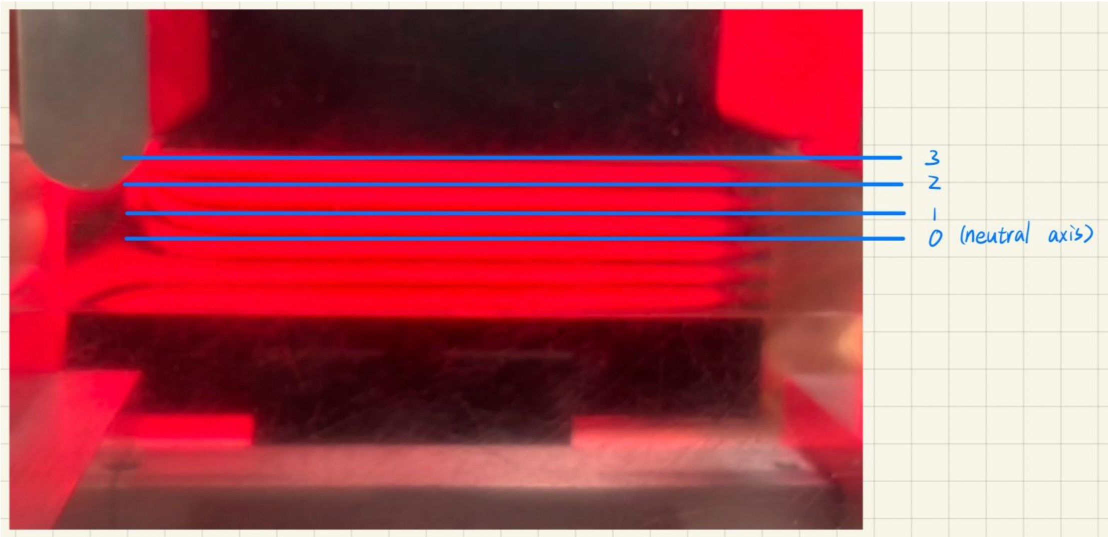

Figure 3. Neutral axis and stripe numbers for $m = 3\ kgw, l = 2.5\ cm$

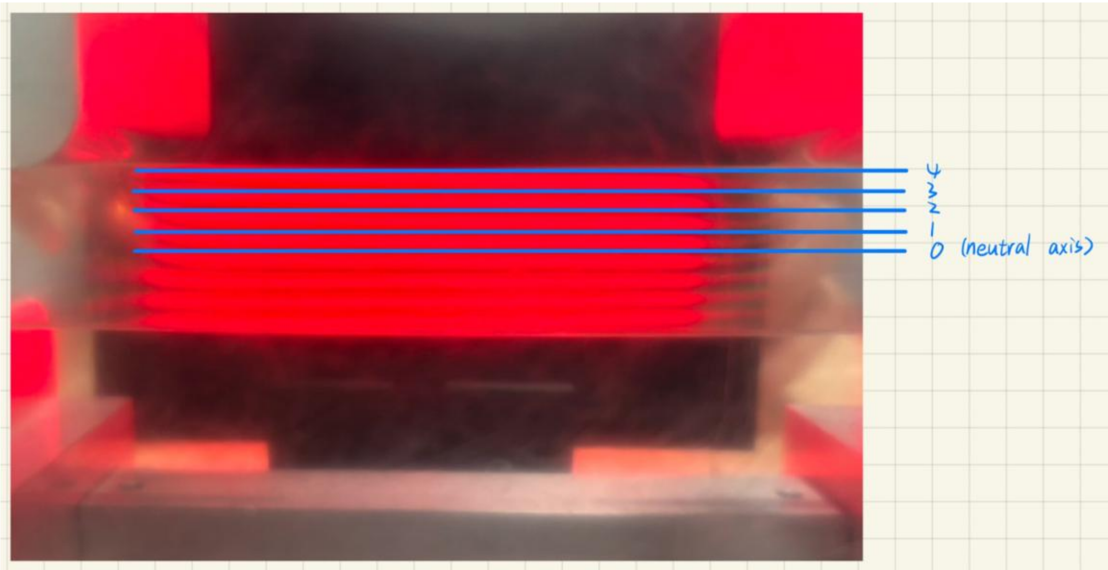

Figure 4. Neutral axis and stripe numbers for $m = 3.5\ kgw, l = 3.1\ cm$

(b) .

Formula 13: $\sigma_1 = \frac{12Ply}{hw^3} = \frac{Nf}{h}, f = \frac{12Ply}{Nw^3}$

For figure 4 in Q1(a):

$2P = 3.5 \times 6\ kgw = 21\ kgw = 21 \times 9.81 =$

$206.01\ N\ \ (the\ loads\ are\ placed\ on\ the\ 6th\ hole)$

$l = 3.1\ cm = 0.031\ m$

$w = 2.43\ cm = 0.0243\ m$

$y = 0.5 \times w = 0.5 \times 0.0243 = 0.01215\ m$

$h = 0.0056\ m$

$N = 4$

$f = \frac{12Ply}{Nw^3} = \frac{12\times\frac{206.01}{2}\times0.031\times0.01215}{4\times0.0243^3} = 8111.454\ N/m$

Q2:

For $m = 1\ kgw, l = 2\ cm$:

$l = 2\ cm = 0.02\ m$

$2P = 1 \times 6\ kgw = 6\ kgw = 6 \times 9.81 = 58.86\ N$

Theoretical estimation: $\sigma_{max,t} = \frac{6Pl}{hw^2} = \frac{6\times\frac{58.86}{2}\times0.02}{0.0056\times0.0243^2} = 1067999.216\ N/m^2$

Experimental estimation: $\sigma_{max,e} = \frac{N_{surface}f}{h} = \frac{0.5\times8111.454}{0.0056} = 724236.964\ N/m^2$

$Error = \frac{\sigma_{max,e}-\sigma_{max,t}}{\sigma_{max,t}} = \frac{724236.964-1067999.216}{1067999.216} = -32.188\%$

For $m = 2.5\ kgw, l = 1.7\ cm$:

$l = 1.7\ cm = 0.017\ m$

$2P = 2.5 \times 6\ kgw = 15kgw = 15 \times 9.81 = 147.15\ N$

Theoretical estimation: $\sigma_{max,t} = \frac{6Pl}{hw^2} = \frac{6\times\frac{147.15}{2}\times0.017}{0.0056\times0.0243^2} = 2269498.334\ N/m^2$

Experimental estimation: $\sigma_{max,e} = \frac{N_{surface}f}{h} = \frac{1\times8111.454}{0.0056} = 1448473.929\ N/m^2$

$Error = \frac{\sigma_{max,e}-\sigma_{max,t}}{\sigma_{max,t}} = \frac{1448473.929-2269498.334}{2269498.334} = -36.176\%$

For $m = 3\ kgw, l = 2.5\ cm$:

$l = 2.5\ cm = 0.025\ m$

$2P = 3 \times 6\ kgw = 18kgw = 18 \times 9.81 = 176.58\ N$

Theoretical estimation: $\sigma_{max,t} = \frac{6Pl}{hw^2} = \frac{6\times\frac{176.58}{2}\times0.025}{0.0056\times0.0243^2} = 4004997.061\ N/m^2$

Experimental estimation: $\sigma_{max,e} = \frac{N_{surface}f}{h} = \frac{3\times8111.454}{0.0056} = 4345421.786\ N/m^2$

$Error = \frac{\sigma_{max,e}-\sigma_{max,t}}{\sigma_{max,t}} = \frac{4345421.786-4004997.061}{4004997.061} = 8.500\%$